

Phase D3 Final Closure: Twenty-Seven FAI Anti-Patterns in Seven Categories Establishing Complete Anti-Pattern Prior Art for Paper 3 FAI Governance, With Transition to Phase D4 Composition Pairs

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Date: May 15, 2026

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This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

This note is the thirtieth and final note of Phase D3 in the Series D defensive-publication derivation record. Phase D3 (notes D3.01–D3.30, #576–#605) has established a complete anti-pattern prior-art record for Paper 3 FAI governance through twenty-seven named anti-patterns organized into seven categories. Each anti-pattern carries detection criteria, identifies the Paper 3 governance commitment it violates, and locates its intra-Self analog in Series A or Series B where applicable. This final note summarizes what each category established, states Phase D3's prior-art value as a closed taxonomy, and transitions to Phase D4, which covers composition pairs — governance scenarios where two correct Paper 3 commitments interact simultaneously and produce non-obvious requirements. The combined Phase D0 through D3 record (146 notes, #460–#605) constitutes a comprehensive prior-art foundation for Paper 3.

1. Phase D3 in context

Phase D3 is the anti-pattern phase of Series D, the derivation record for Paper 3's inter-Self coordination architecture. Its function within the series is defensive: where Phase D1 established what Paper 3's governance commitments require and Phase D2 decomposed those commitments into operational variants, Phase D3 identifies what failure looks like when the commitments are not honored. A prior-art record without failure-mode documentation leaves a gap that adversarial novelty claims can exploit — the claim that a specific failure mode, and therefore the governance commitment that prevents it, was not previously identified. Phase D3 closes that gap.

The phase opened with note D3.01 (note #576), which introduced the twenty-seven-anti-pattern taxonomy organized into seven categories and specified the seven-element structure all standalone anti-pattern notes would follow: name, definition, governance commitment violated, detection criteria, intra-Self analog, co-occurrence risk, and operational consequence. Notes D3.02 through D3.29 covered all twenty-seven anti-patterns plus one cross-category co-occurrence analysis. This note closes the phase.

2. Complete taxonomy by category

Category 1 — Lifecycle failures (D3.02–D3.04). Anti-pattern 1 (Ungoverned Construction), Anti-pattern 2 (Ungoverned Dissolution), and Anti-pattern 13 (Silent Offboarding) address governance failures at the beginning and end of a shared substrate’s operational life. These three notes established that Paper 3’s shared-substrate claim requires human governance not only during active operation but at the structural moments when a substrate comes into existence and when it is dissolved or a participating Self departs. The analogs to Paper 2’s B3.10 and B3.11 were identified, grounding the category in the intra-Self lifecycle record.

Category 2 — Conflict handling failures (D3.05–D3.08). Anti-pattern 3 (Silent Conflict Collapse), Anti-pattern 11 (Unresponsive Escalation), Anti-pattern 12 (Informal Resolution Rules), and Anti-pattern 14 (Competitive Intelligence Misrouting) address the three-tier conflict-handling mechanism that Paper 3’s Claim 3 specifies. These four notes established that the preserve tier, the resolve tier, and the escalate tier each fail in distinct ways, and that failure in the tier-transition logic is analytically distinct from failure within a tier. The competitive-intelligence misrouting anti-pattern established that the cooperative/competitive variant distinction has its own conflict governance failure mode.

Category 3 — Evolution feed failures (D3.09–D3.12). Anti-pattern 4 (Automatic DNA Absorption), Anti-pattern 5 (Non-Attributed Ingestion), Anti-pattern 15 (Vertical Propagation Break), and Anti-pattern 16 (Unconfigured Proposing Substrate) address Paper 3’s four-locus evolution-feed claim. These four notes established that each locus of the evolution feed is a distinct failure site. Anti-pattern 16 was additionally classified as an optimization failure, establishing that evolution-feed governance encompasses not only correctness but productive use of the mechanism.

Category 4 — Configuration failures (D3.13–D3.17). Anti-pattern 6 (Implicit Configuration), Anti-pattern 9 (Pseudo-Selective Merge), Anti-pattern 17 (N-Blind Configuration), Anti-pattern 18 (Forgotten Standing Configuration), and Anti-pattern 19 (Time-Pressure Governance Bypass) address Paper 3’s configuration-as-substrate-content claim. These five notes established that configuration governance failures are structurally more varied than failures in other categories, including failures at initial configuration, ongoing maintenance, population-size dimension, merge-scope specification, and governance bypass under external pressure. The last of these established that governance bypass under urgency is a named failure mode, not merely a deployment risk.

Category 5 — Perimeter and content failures (D3.18–D3.21). Anti-pattern 7 (Home Perimeter Erosion), Anti-pattern 8 (Exchange Bounding Violation), Anti-pattern 10 (Shallow-Reference Merge), and Anti-pattern 20 (FAI-Oblivious Restructuring) address boundary and content-integrity requirements. These four notes established that boundary governance failures operate at two levels — the perimeter of each Self’s home substrate and the exchange-bounding rule for what content crosses the perimeter during an FAI event — and that content governance failures include both merge depth and organizational change that disrupts the FAI relationship without governing that disruption.

Category 6 — Governance quality failures (D3.22–D3.25). Anti-pattern 21 (Governance Theater), Anti-pattern 22 (Black Box Shared Substrate), Anti-pattern 23 (Authorization Chain Gap), and Anti-pattern 24 (Automated Governance) address failures of governance quality rather than failures in specific governance mechanisms. These four notes established a severity spectrum: Governance Theater at the compliance-form end; Automated Governance — the substitution of algorithmic governance for human governance — as the most severe anti-pattern in the full taxonomy. The severity ordering across the category was explicitly established.

Category 7 — Population-scope failures (D3.26–D3.28). Anti-pattern 25 (Topology Lock-In), Anti-pattern 26 (Participation Mode Gaming), and Anti-pattern 27 (Forced Propagation) address Paper 3’s population-scale collective evolution claim. These three notes, written in the calibrated-humility register appropriate to Paper 3’s population-scope commitments, established that population-level governance failures are structurally distinct from cell-level or pair-level failures: they emerge from the accumulated behavior of many FAI events and may not be detectable at the level of any individual event.

Cross-category co-occurrence analysis (D3.29). The penultimate Phase D3 note identified four co-occurrence patterns across the twenty-seven anti-patterns, named the three most dangerous combinations, and specified a detection priority order. This analysis established that the anti-pattern taxonomy has internal structure beyond category membership — certain anti-patterns reliably co-occur, and their combination produces governance degradation that exceeds the sum of their individual effects.

3. Phase D3’s prior-art value

Phase D3’s primary contribution is a closed anti-pattern taxonomy for Paper 3 FAI governance. The taxonomy is closed in the sense that it was constructed by systematic derivation from Paper 3’s six claims: every anti-pattern corresponds to a governance commitment violation traceable to a specific claim, and every category corresponds to a governance domain that one or more claims address. The systematic construction, not merely the count of anti-patterns, is what produces the prior-art coverage.

The prior-art implication is precise: any adversarial claim that a specific FAI governance failure mode is novel must locate that failure mode outside all twenty-seven named anti-patterns and their associated detection criteria. The detection criteria are the operative constraint — they specify the observable conditions under which each anti-pattern is present, not merely a conceptual definition. A failure mode that matches any anti-pattern's detection criteria falls within that anti-pattern's prior-art coverage regardless of the terminology used to describe it.

The intra-Self analog structure reinforces the coverage. Each anti-pattern, where applicable, was linked to its Series A or Series B equivalent, placing the Paper 3 failure mode within the broader CKS failure-mode record. This cross-series linkage means that novelty arguments must also address whether the claimed novel failure mode is merely an inter-Self extension of an already-established intra-Self anti-pattern.

4. The 146-note milestone

This note completes the fourth phase of Series D and brings the combined Phase D0 through D3 note count to 146 (#460–#605). The milestone is worth naming precisely: Phase D0 established the six claim-anchor notes for Paper 3; Phase D1 decomposed each claim into its foundational sub-commitments (~30 notes); Phase D2 decomposed each sub-commitment into operational variants (~80 notes); Phase D3 documented the twenty-seven anti-patterns in seven categories (30 notes). Together, these four phases constitute a comprehensive prior-art foundation — claim coverage, sub-commitment coverage, operational coverage, and failure-mode coverage — for Paper 3's governance architecture.

Phases D4 through D6 will add composition pairs, operational tests, and boundary cases. That additional work will deepen and stress-test the record. But the foundation is now complete in the sense that matters for defensive publication: every governance commitment Paper 3 makes has been established, decomposed, operationalized, and documented with failure modes. The foundation does not require Phases D4–D6 to be defensible; those phases add analytical value beyond the baseline.

5. Transition to Phase D4: composition pairs

Phase D4 (~30 notes, #606–#635) covers composition pairs. The functional distinction from Phase D3 is precise and should be stated directly: anti-patterns document what happens when governance commitments are violated; composition pairs document what happens when governance commitments are honored correctly but applied simultaneously to two interacting commitments. The failure is not present. The governance is operating as specified. The analytical question is different: what does correct governance of commitment A require when commitment B is also in force, and how does the interaction between A and B produce requirements that neither commitment generates on its own?

Phase D4 will cover composition pairs within Paper 3 — for example, the shared-substrate commitment combined with the FAI commitment, or the three-tier conflict-handling commitment combined with the four-locus evolution-feed commitment — and composition pairs bridging Papers, where a Paper 1 or Paper 2 commitment interacts with a Paper 3 commitment at the inter-Self scope. The bridging pairs are analytically distinct from the within-Paper pairs: they produce requirements that span the intra-Self and inter-Self scopes simultaneously, a form of interaction that neither paper addresses in isolation.

The composition-pair format will follow the same structural discipline as Phase D3: each note will identify the two commitments being composed, state the interaction requirements produced, and locate any prior-art coverage or gap in the existing record. The transition from Phase D3 to Phase D4 is therefore not a change in analytical rigor but a change in the analytical question being asked — from what breaks when governance fails to what is required when governance succeeds.

6. Conclusion

Phase D3 (notes D3.01–D3.30, #576–#605) has established a complete anti-pattern prior-art record for Paper 3 FAI governance through twenty-seven named anti-patterns in seven categories. The taxonomy covers all six Paper 3 claims, provides detection criteria for each failure mode, identifies intra-Self analogs across the CKS series record, and includes cross-category co-occurrence analysis. Any adversarial novelty claim about a FAI governance failure mode must address this record.

The combined Phase D0 through D3 record (146 notes) constitutes a comprehensive prior-art foundation for Paper 3. Phase D4 opens with a functionally distinct analytical agenda: not what breaks, but what non-obvious governance requirements emerge when two correct commitments are applied simultaneously. The foundation is complete. Phase D4 begins.

Source papers

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

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How to cite this note

Li, W. (2026). *Phase D3 Final Closure: Twenty-Seven FAI Anti-Patterns in Seven Categories Establishing Complete Anti-Pattern Prior Art for Paper 3 FAI Governance, With Transition to Phase D4 Composition Pairs*. May 15, 2026. ORCID: 0009-0004-8065-3235.